

CERTIFICATION PROGRAM IN INTRODUCTION TO GENERATIVE AI

Course Code	BCA-GAI401	CIE Marks	25
Teaching Hours/Week (L:P:SDA)	1:2:0	SEE Marks	25
Total Hours of Pedagogy	30	Total Marks	50
Credits	00	Exam Hours	03

Commandments adopted:

- C1 - Take full ownership of your learning and career
- C3 - Respect infrastructure and value every opportunity
- C4 - Commit to continuous enrichment and excellence

Course Objectives

- **Evaluate Generative AI outputs critically:**

Students will learn to assess AI-generated content for accuracy, relevance, limitations and responsible use in academic and professional contexts.

- **Use and evaluate Generative AI tools:**

Students will gain hands-on experience with Generative AI tools, local LLMs (via Ollama) and simple AI workflows for practical everyday tasks.

- **Apply Prompt Engineering techniques:**

Students will learn to craft effective prompts for academic, professional and software-development tasks.

- **Develop understanding of LLMs and Transformers:**

Students will build practical knowledge of Large Language Models, tokens, context and Transformers through hands-on experimentation.

- **Understand fundamental AI concepts:**

Students will learn the fundamental concepts and practical applications of Artificial Intelligence, Machine Learning, Deep Learning and Generative AI.

- **Drive innovation:**

To use Generative AI techniques to explore practical solutions, support informed decisions, improve productivity, and develop innovative solutions to practical problems.

Course prerequisites

- No prior Generative AI experience is required.
- Basic programming familiarity is helpful but not mandatory.
- Basic logical reasoning.
- Basic internet and browser usage.
- Basic computer literacy.

System Requirements

- Recommended: 16 GB RAM, 30-50 GB available SSD storage.
- Minimum: 8 GB RAM, 20 GB available SSD storage, operating system capable of running Ollama, Python 3.x recommended.

Note:

Local model performance depends on model size and available system resources. Smaller models should be used on low-resource systems. The minimum configuration does not guarantee efficient execution of every available language model.

Certifications

- IBM SkillsBuild — Getting Started with Generative AI (10 hours)
- IBM SkillsBuild — Generative AI Essentials: Using LLMs to Work with Data (27 hours)
- Microsoft Learn — Generative AI / AI Fluency learning credentials (12 hours)
- Google Cloud — Generative AI Leader (18 hours)
- IBM Generative AI Engineering Professional Certificate (20 hours)

Introduction to Artificial Intelligence and Generative AI

- What is Artificial Intelligence?
- AI around us - examples in daily life
- AI in education and learning
- Machine Learning
- Deep Learning
- Generative AI
- Traditional AI vs Generative AI
- Text generation
- Image generation
- Audio and speech generation
- AI assistants and chatbots
- Large Language Models - what is an LLM?
- Basic idea of model training
- Training vs inference
- Context
- Tokens - introductory concept
- AI capabilities and limitations
- Hallucinations
- Bias
- Privacy
- Responsible AI
- Cloud AI vs Local AI - introducing Ollama
- Running a small local model with Ollama

Basic Ollama Practical:

- Students check Ollama installation (ollama --version) and run a small local model.
- Students ask simple questions and observe local AI responses.
- Students compare local AI output with cloud AI (e.g., ChatGPT).

Module End Assignment ::

"AI Around Me": ‘

Students identify and explain 10 real-world AI use cases.

Module End Project ::

Create an AI Use-Case Portfolio.

Module End Test ::

AI and Generative AI fundamentals test.

Recommended Reading ::

Microsoft Learn - Generative AI learning paths

(learn.microsoft.com)

Microsoft Generative AI for Beginners

(github.com/microsoft/generative-ai-for-beginners)

LLMs, Tokens and Transformers - How Generative AI Works

- What is an LLM?
- Language models
- Training data
- Model training - simple explanation
- Inference
- Tokens
- Why text is converted into tokens
- Tokenization - simple demonstration
- Context window
- Context and conversation
- Token limitations
- Token usage awareness
- What is a Transformer?
- Why Transformers became important
- Basic Transformer intuition
- Attention - intuitive explanation only
- How context influences output
- Input -> processing -> output
- Model limitations
- Hallucinations - why LLMs can sound confident but be wrong

Module End Assignment ::

"Inside an LLM - Explain It Simply":

Students explain LLM, Token, Context, and Transformer using their own words and an everyday analogy.

Module End Project ::

Create a one-page "How Generative AI Works" learning poster/document.

Module End Test ::

LLM + Token + Transformer test.

Recommended Reading ::

- IBM SkillsBuild - Generative AI resources (skillsbuild.org)
- OpenAI API documentation (platform.openai.com/docs)

Prompt Engineering and Effective AI Use

- What is Prompt Engineering?
- Poor prompts vs better prompts
- Role
- Context
- Task
- Constraints
- Examples
- Output format
- Zero-shot prompting
- Few-shot prompting
- Structured prompts
- Prompt decomposition
- Iterative prompting
- Asking for explanations
- Asking for comparisons
- Asking for summaries
- Asking for tables
- Asking for structured output (JSON)
- Prompt refinement
- Evaluating AI output
- Hallucination detection
- Fact verification
- Prompt injection awareness
- Privacy and sensitive information
- Temperature - practical experiment

Temperature Experiment:

Use the same prompt with different temperature settings (e.g., 0.2, 0.5, 0.8, 1.0).

Students record: Prompt, Temperature, Output, Observation.

Students observe how temperature affects creativity, predictability and variation.

Module End Assignment ::

Prompt Improvement Challenge: Students improve 10 poor prompts into well-structured prompts.

Module End Project ::

BCA Student Prompt Handbook containing minimum 20 practical prompts covering: Learning, Coding, SQL, Resume, Interview, Email, Presentation, Debugging, and Documentation.

Module End Test ::

Prompt Engineering Practical Challenge.

Recommended Reading ::

Python official documentation (docs.python.org)
REST API learning resources
JSON documentation (json.org)

AI Agents, APIs and AI-Powered Applications

- **AI FOR LEARNING:**

- Explain concepts simply
 - Create examples and quizzes
 - Summarize notes and study plans

- **AI FOR SOFTWARE DEVELOPMENT:**

- Code explanation and debugging
 - Error explanation and code improvement
 - Test-case generation and documentation
 - SQL, Python, HTML/CSS assistance
 - Verifying AI-generated code

- **AI FOR PROFESSIONAL WORK:**

- Email drafting and resume improvement
 - Interview preparation
 - Documentation and presentation planning

- **AI OUTPUT EVALUATION:**

- Accuracy, relevance, completeness
 - Hallucinations and bias
 - Verification and human judgment
 - GENERATE -> CHECK -> VERIFY -> CORRECT -> USE

- **AI AGENTS:**

- What is an AI Agent?
 - AI assistant vs AI agent
 - Agent loop: GOAL -> PLAN -> USE TOOL -> OBSERVE -> RESPOND
 - Human-in-the-loop
 - Agent limitations and risks

Module End Assignment ::

AI Output Evaluation Challenge:

Students evaluate 10 AI-generated responses for accuracy, hallucinations and bias.

Module End Project ::

Create an AI Productivity Toolkit:

A collection of prompts and workflows for learning, coding and professional tasks.

Module End Test ::

Practical AI-use + output-evaluation test.

Recommended Reading ::

Deep Learning by Ian Goodfellow, Yoshua Bengio,
Aaron Courville

Artificial Intelligence Engines by James Stone

Module 5:	(6 hours)
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Capstone Project - Build a Small AI Assistant

- **Students choose ONE project:**
 - Local AI BCA Study Assistant
 - Local AI College FAQ Assistant
 - Local AI Interview Assistant
 - Local AI Resume Helper
 - Local AI Notes Summarizer

- **Capstone requirements:**
 - Identify the problem and target user
 - Define inputs and expected outputs
 - Create structured prompts
 - Test different prompts and temperature settings
 - Evaluate AI responses and identify incorrect outputs
 - Improve prompts based on evaluation
 - Build a small working prototype or guided AI workflow
 - Document limitations and responsible-AI considerations
 - Demonstrate the solution

- **Capstone deliverables:**
 - Working prototype / AI workflow
 - Problem statement and target user
 - Prompt design and sample inputs/outputs
 - Temperature experiment results
 - Screenshots and output evaluation
 - Corrections and improvements
 - Limitations and responsible AI considerations
 - README and short presentation

Module End Assignment ::

College AI Application Proposal:

Students explain the problem, target user, chosen AI approach, sample prompts, and expected workflow.

Module End Project ::

Complete Small AI Assistant using Ollama/local LLM as described above.

Recommended Reading ::

- Human Compatible by Stuart Russell
- Architects of Intelligence by Martin Ford

Module End Test ::

Mini Project Demonstration + Viva.

Certifications

On successful completion of all five modules, submission of the prescribed module-end assignments and project deliverables, successful completion of the capstone project, and fulfillment of the assessment requirements, students will be awarded the "Certificate Program in Introduction to Generative AI" Course Completion Certificate.

Course Outcomes (CO)

CO5: Create, demonstrate and evaluate a small Generative AI-based solution for a practical user problem.

CO4: Use and evaluate Generative AI tools, local LLMs and simple AI workflows while considering accuracy, limitations and responsible use.

CO3: Apply Prompt Engineering techniques to solve academic, professional and software-development tasks.

CO2: Explain the basic concepts of Large Language Models, tokens, context and Transformers.

CO1: Explain the fundamental concepts and practical applications of Generative AI.

CO-PO Mapping

PO6: Professionalism & Ethics

PO5: Lifelong Learning

PO4: Communication

PO3: Practical & Tool Usage

PO2: Problem Analysis

PO1: Domain Knowledge

Mapping scale: 1 = Low, 2 = Medium, 3 = High correlation.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	2	1	1	2	2
CO2	3	2	2	1	2	2
CO3	3	3	3	2	3	2
CO4	3	3	3	2	3	3
CO5	3	3	3	3	3	3

Reference Books

"Co-Intelligence: Living and Working with AI" - Ethan Mollick

"Generative AI: A Short Introduction" - Egidio P. Costa

"Python Crash Course" - Eric Matthes

"Artificial Intelligence: A Guide for Thinking Humans" - Melanie Mitchell

Reference E-Resources

Microsoft Learn - Generative AI (learn.microsoft.com)

Microsoft Generative AI for Beginners (github.com/microsoft/generative-ai-for-beginners)

IBM SkillsBuild - Generative AI (skillsbuild.org)

Python Documentation (docs.python.org)

OpenAI API Documentation (platform.openai.com/docs)

REST API Learning Resources (restfulapi.net)

JSON Documentation (json.org)

Ollama Official Documentation (ollama.com)

Reference Video Resources

Generative AI Fundamentals - Microsoft Learn

Introduction to Generative AI - freeCodeCamp (YouTube)

Prompt Engineering Tutorial - freeCodeCamp (YouTube)

Large Language Models Explained - Microsoft (YouTube)

AI Agents and APIs - IBM Technology (YouTube)

Google Skills for all

Hack to Skills by google